

Same Pipeline, Drawn Opposite Ways:

A Note on the f_0 / Normalizing-Flow / Hamiltonian-Flow Picture

Verdict

You’re both drawing the same pipeline, just in opposite directions — and that’s allowed, because every map involved is invertible. So neither of you is “wrong about the mechanics.” But on the *interpretation* of what the objects mean and what the deliverable is, you are more correct than your supervisor on two specific points, and he is right on one. Let me separate them.

The one pipeline, written precisely

Data generation (forward).

$$f_0^{\text{true}}(z_0) \xrightarrow{\text{sample}} \{z_0^i\} \xrightarrow{\text{HF}(h_{\text{true}})} \{z_1^i\} \quad (\text{the observed data}).$$

Inference. The likelihood of the data under a model (h, θ) is evaluated as

$$z_1 \xrightarrow{\text{HF}^{-1}(h)} z_0(h) \xrightarrow{g_\theta} \tilde{z}_0 \sim \mathcal{N}(0, I),$$

$$f_{\text{model}}(z_1) = \underbrace{\mathcal{N}(g_\theta(z_0(h))) \left| \det \frac{\partial g_\theta}{\partial z_0} \right|}_{\bar{f}_\theta(z_0)} \cdot \underbrace{\left| \det \frac{\partial \text{HF}^{-1}}{\partial z_1} \right|}_{= 1 \text{ (symplectic)}}.$$

Your supervisor’s arrows go up ($z_1 \rightarrow z_0 \rightarrow \tilde{z}_0$). That is exactly the normalizing / likelihood-evaluation direction. It’s correct.

Your arrows emphasize the generative / interpretation direction ($\tilde{z}_0 \rightarrow z_0$, giving $\bar{f}_\theta(z_0)$ to compare against f_0^{true}). Also correct.

They’re the same diagram because g_θ and HF are both invertible. So the “disagreement” is not about the code — it’s about semantics.

Where you are right

1. **The standard Gaussian base is a tool, not a distribution in the chain.** Your supervisor gives the top level the status of a reconstructed DF, $\hat{f}_0(\tilde{z}_0)$. But $\tilde{z}_0$ is the flow’s fixed latent reference measure — it is $\mathcal{N}(0, I)$ by construction, identical for every dataset and every θ . There is nothing to “compare” it to. The object you actually compare to f_0^{true} lives in the z_0 space and is $\bar{f}_\theta(z_0)$. So promoting the latent Gaussian to a named deliverable is the misleading part of 7166. You’re right to demote it.

2. **HF and NF are not the same kind of transformation.** This is the deeper point. The NF (g_θ) is a general diffeomorphism with a nontrivial Jacobian — it genuinely reshapes the density (that’s how it turns a Gaussian into a double-Gaussian, wings, etc.). Hamiltonian back-integration is symplectic $\Rightarrow$ unit Jacobian $\Rightarrow$ Liouville: it does *not* reshape f . It just re-coordinatizes it,

$$f_0(z_0) = f_1(z_1) \quad \text{along an orbit.}$$

So “ $f_1(z_1) \rightarrow f_0(z_0)$ ” as if HF transforms one density into a different one blurs the physics. HF moves points; the DF value rides along unchanged. Only the NF is doing distribution learning.

Where your supervisor is right (and you should concede)

The whole thing can legitimately be read as one composition of invertible change-of-variables maps, and the object compared to truth does live in the z_0 space — which is what he means by “the outcome is f_0 compared to f_{true} .” That statement is fine. His picture is a valid shorthand; it just over-dignifies the latent Gaussian and glosses the HF-vs-NF asymmetry.

Two small precisions to your own wording

- “ $\{z_0^i(h)\} \xrightarrow{\text{NF}} \bar{f}_\theta$ ” makes it sound like the NF maps points into a function. More precisely: $\bar{f}_\theta$ is a fixed parametric density family, and MLE picks θ (jointly with h) so the back-integrated cloud $\{z_0^i(h)\}$ has maximum likelihood under $\bar{f}_\theta$. It’s density estimation, not a deterministic point $\rightarrow$ function map.
- Say explicitly that the fit is joint in (θ, h) — which you do — because that’s what makes it a physics measurement rather than pure density estimation.

The point neither diagram makes (the actual crux)

Why is h identifiable at all? Because h enters only through $\text{HF}^{-1}(h)$. The correct h is the one that back-integrates today’s data into a z_0 cloud that is consistent with the assumed/true initial DF. That’s the whole physical content — and, as your own results show, h enters the potential only anharmonically, so this constraint is weak (flat/shallow NLL in h unless the DF is informative). If you want a diagram that settles the argument, put the emphasis there: the NF learns the shape, and h is whatever makes the back-integrated shape self-consistent with f_0^{true} .

Bottom line

Same pipeline, drawn opposite ways. Your framing is the more physically faithful one — Gaussian = tool, HF = Liouville relabeling, goal = recover f_0^{true} and h jointly. His “chain of change-of-variables” view is legitimate but shouldn’t treat the latent Gaussian as an output or HF as a density-resaping step.